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Lower Limb vessels

Dr Yasser Al Harbi

Lecture Objectives

Learning Objectives

Detailed anatomical organisation of the blood vessels of the lower limb; their major course, distribution and areas of supply/drainage

Aim

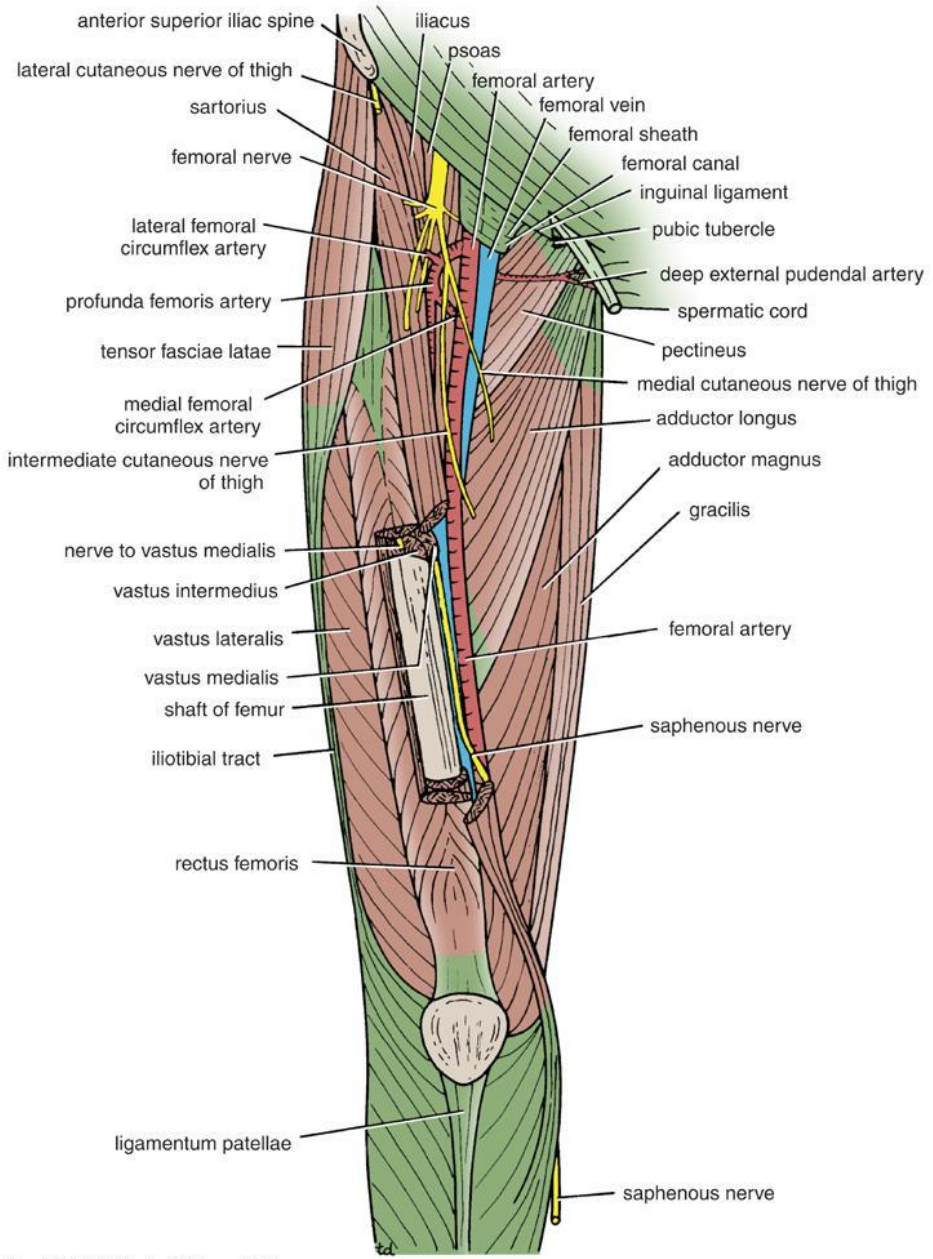
To present an overview of the vessels of the lower limb

Content

- The lymphatic vessels of the lower limb.
- The major arteries of the lower limb.
- The deep and superficial veins of the lower limb.
- The topographical relationships of the arteries and veins of the lower limb.

Femoral Artery

- Its entrance to the thigh
  - Position
    - Midway between ASIS and pubic symphysis
  - Femoral sheath
- Relations
  - Sartorius
  - Iliopsoas & adductor muscles
  - Femoral vein and nerve
- Exit to popliteal region
  - Adductor hiatus



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Femoral Artery

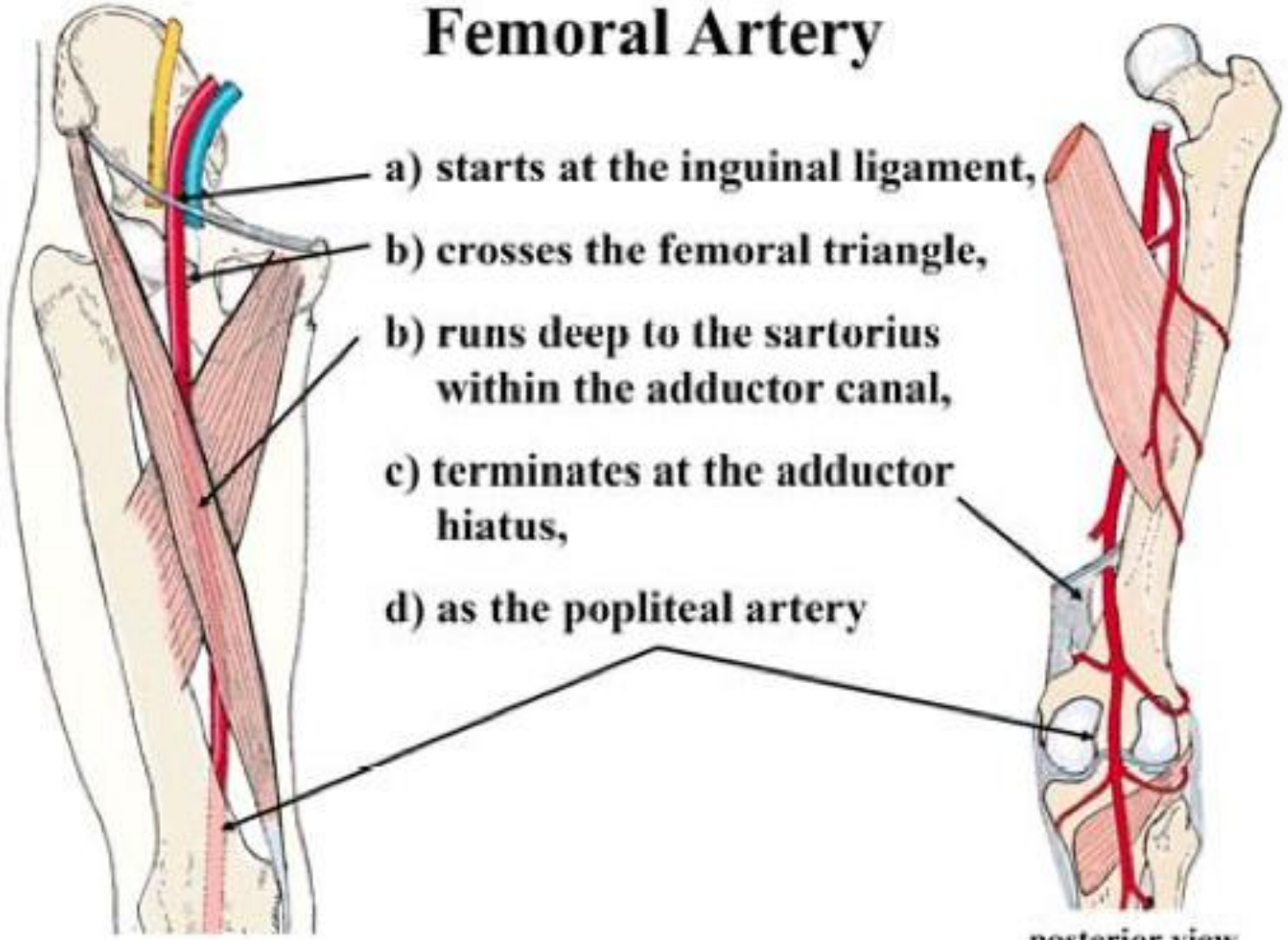
a) starts at the inguinal ligament,

b) crosses the femoral triangle,

b) runs deep to the sartorius within the adductor canal,

c) terminates at the adductor hiatus,

d) as the popliteal artery



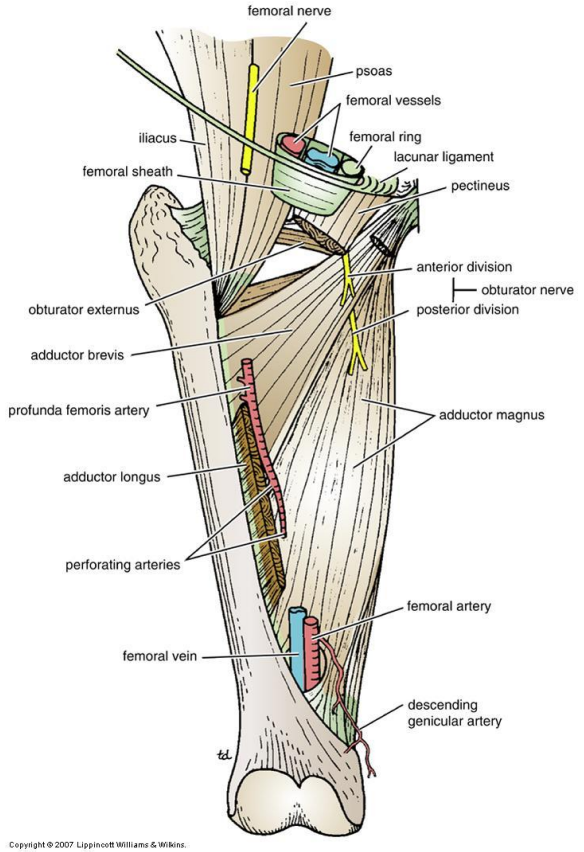
posterior view

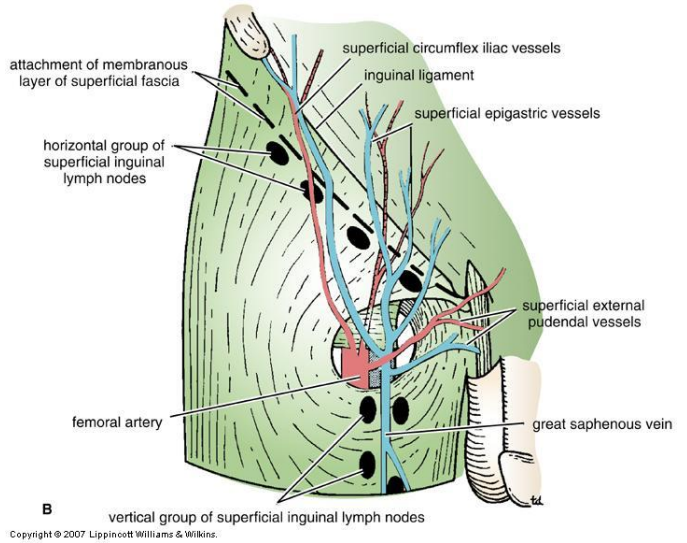
Femoral Artery: Branches

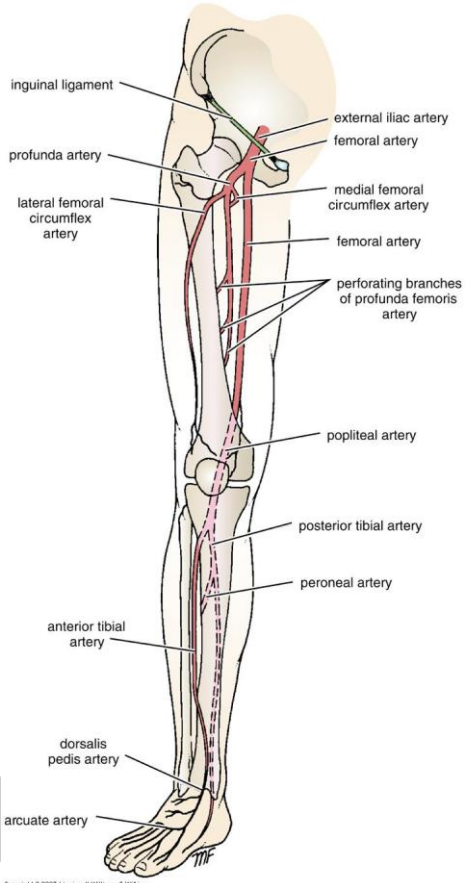
- Superficial circumflex iliac a. (ASIS)
- Superficial epigastric a. (Lower abdominal wall)
- Superficial external pudendal a. (genital skin)
- Deep external pudendal a. (genital skin)
- Profunda femoris a. (deep a. of thigh) (thigh region)

Deep to the adductor longus ➢

- Medial femoral circumflex a.
- Lateral femoral circumflex a.
- 1-4 perforating aa.
- Descending genicular a. (knee joint)





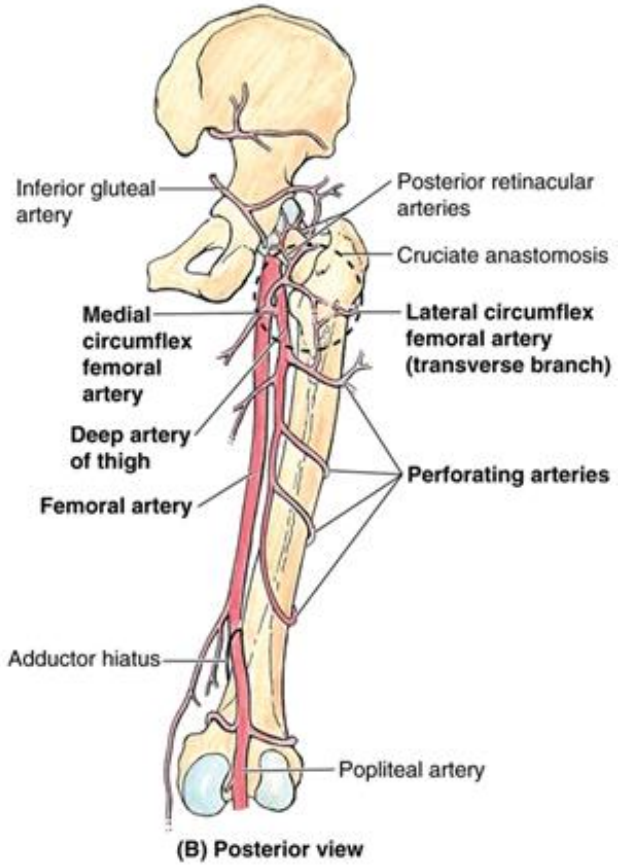


Trochanteric anastomosis

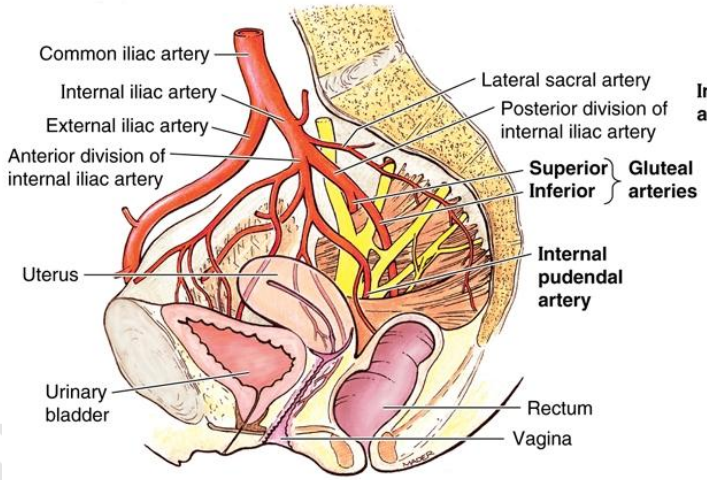
- Superior gluteal a.
- Inferior gluteal a.
- Medial femoral circumflex a.
- Lateral femoral circumflex a.

Cruciate anastomosis

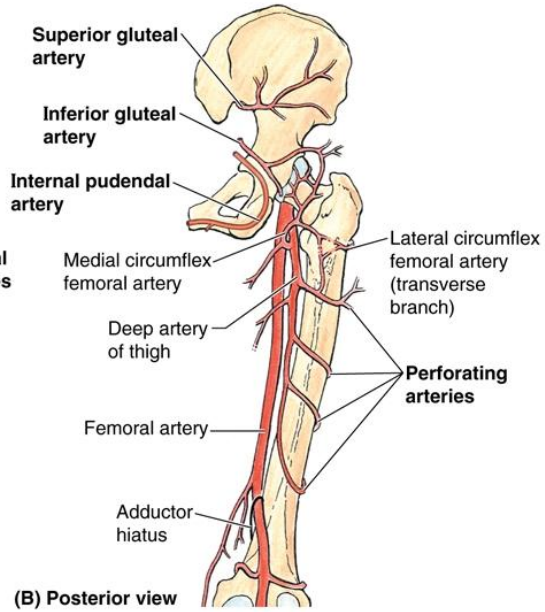
- Inferior gluteal a.
- Medial femoral circumflex a.
- Lateral femoral circumflex a.
- 1st perforating a. from profunda



(B) Posterior view

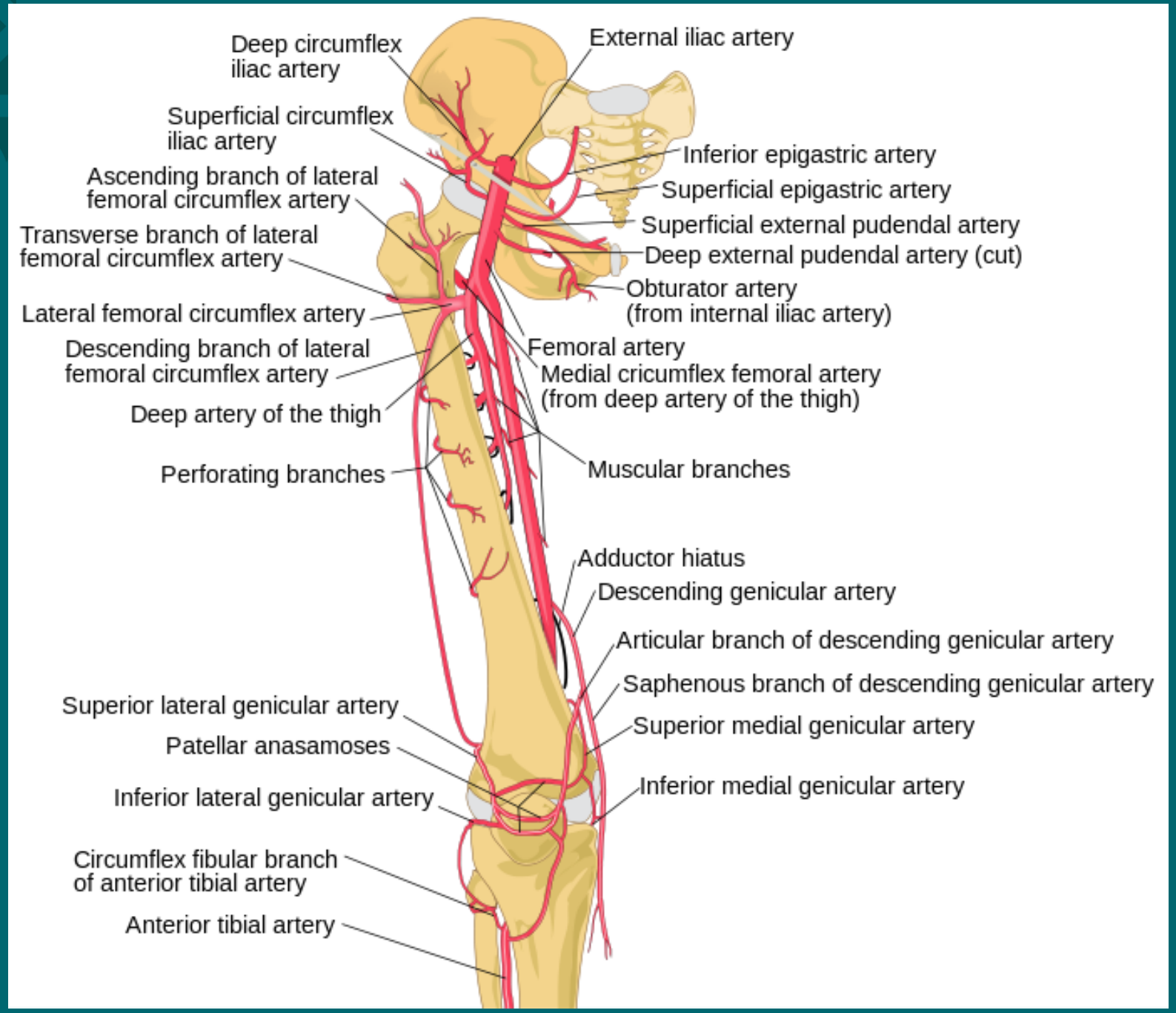


(A) Medial view of right hemipelvis



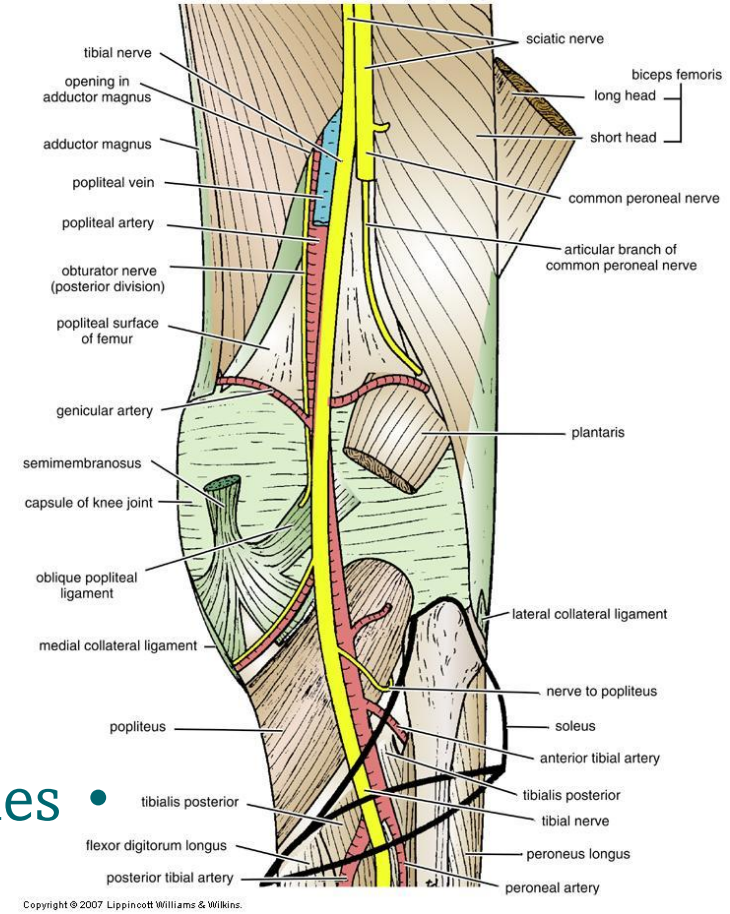
(B) Posterior view

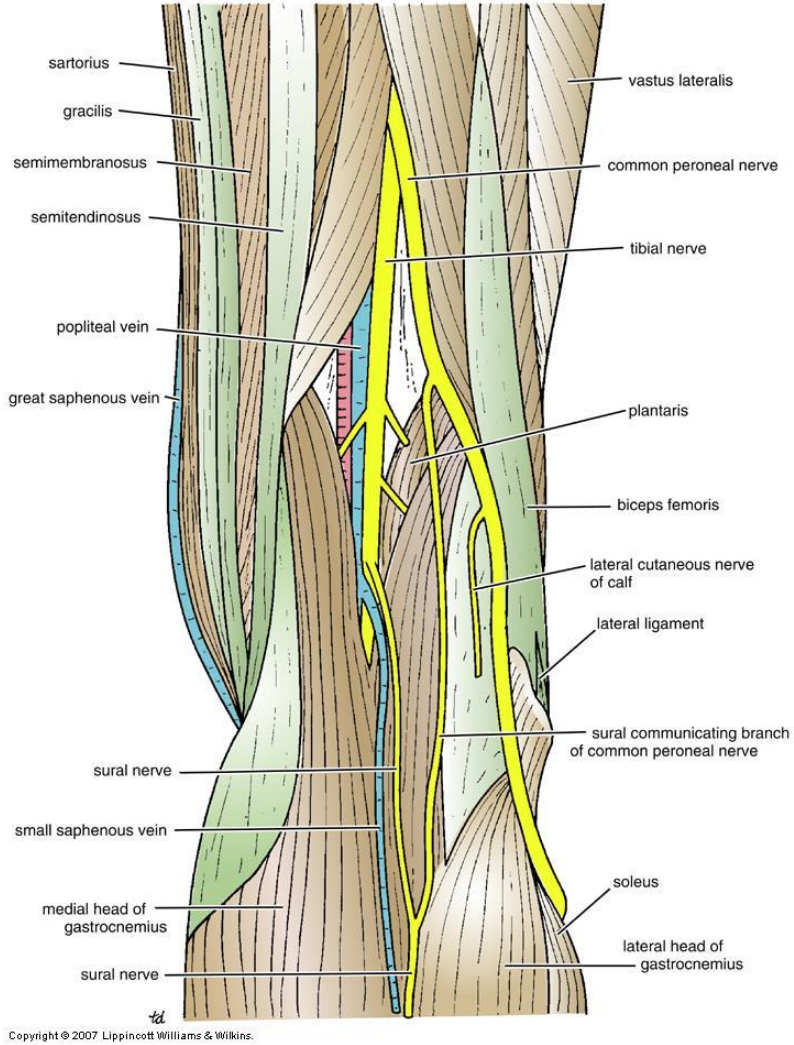
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Deep circumflex iliac artery
External iliac artery
Superficial circumflex iliac artery
Inferior epigastric artery
Ascending branch of lateral femoral circumflex artery
Superficial epigastric artery
Transverse branch of lateral femoral circumflex artery
Superficial external pudendal artery
Deep external pudendal artery (cut)
Obturator artery (from internal iliac artery)
Lateral femoral circumflex artery
Femoral artery
Descending branch of lateral femoral circumflex artery
Medial cricumflex femoral artery (from deep artery of the thigh)
Deep artery of the thigh
Muscular branches
Perforating branches
Adductor hiatus
Descending genicular artery
Articular branch of descending genicular artery
Saphenous branch of descending genicular artery
Superior lateral genicular artery
Superior medial genicular artery
Patellar anasamoses
Inferior medial genicular artery
Inferior lateral genicular artery
Circumflex fibular branch of anterior tibial artery
Anterior tibial artery

Popliteal Artery

- Position •
- Relations •
- Branches •
  - Muscular branches •
  - Articular (genicular) branches •
  - Terminal branches •
    - Anterior tibial a. •
    - Posterior tibial a. •





Arteriograms

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Femoral artery and its branches

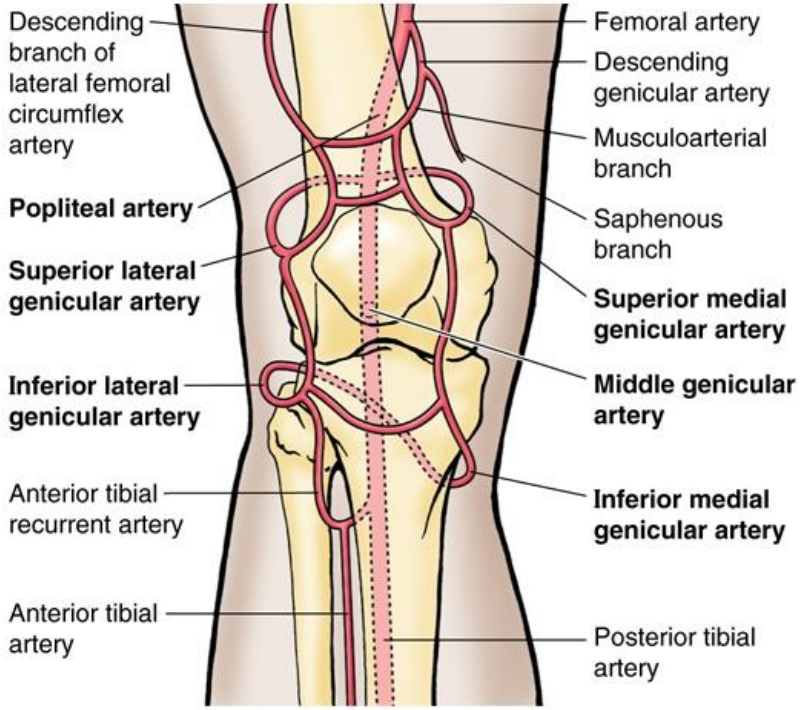


Popliteal artery and its branches



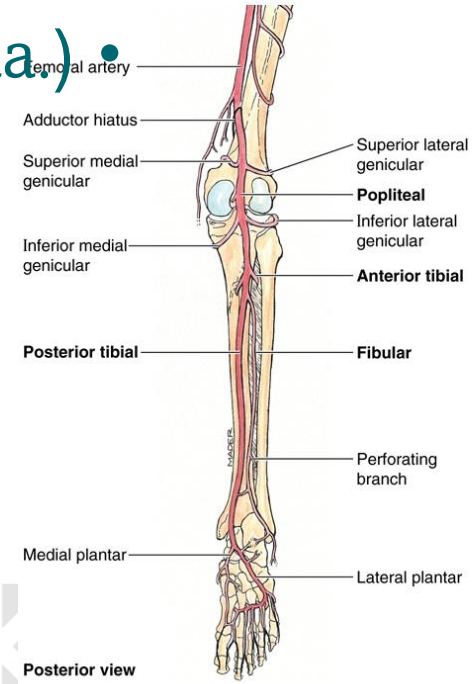
Genicular Anastomosis Around Knee Joint

- Femoral a.
  - Descending genicular a.
  - Descending branch of lateral femoral circumflex a.
- Popliteal a. (genicular aa.)
  - Superior medial & lateral
  - Middle
  - Inferior medial & lateral
- Anterior tibial a.
  - Anterior tibial recurrent a.



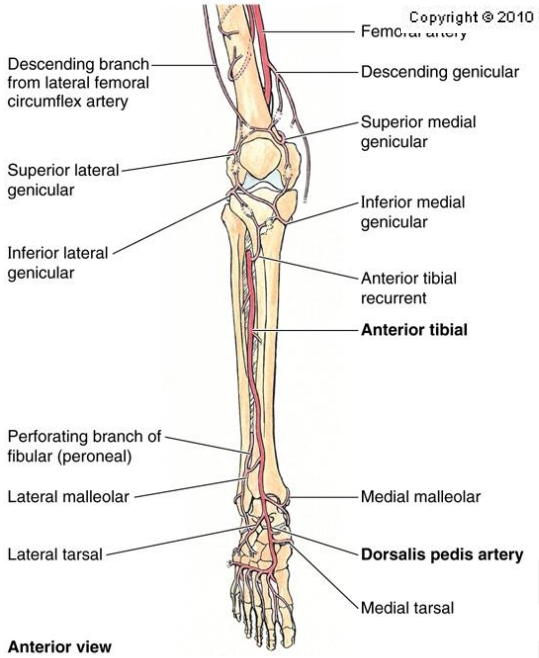
Anterior view

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Posterior view

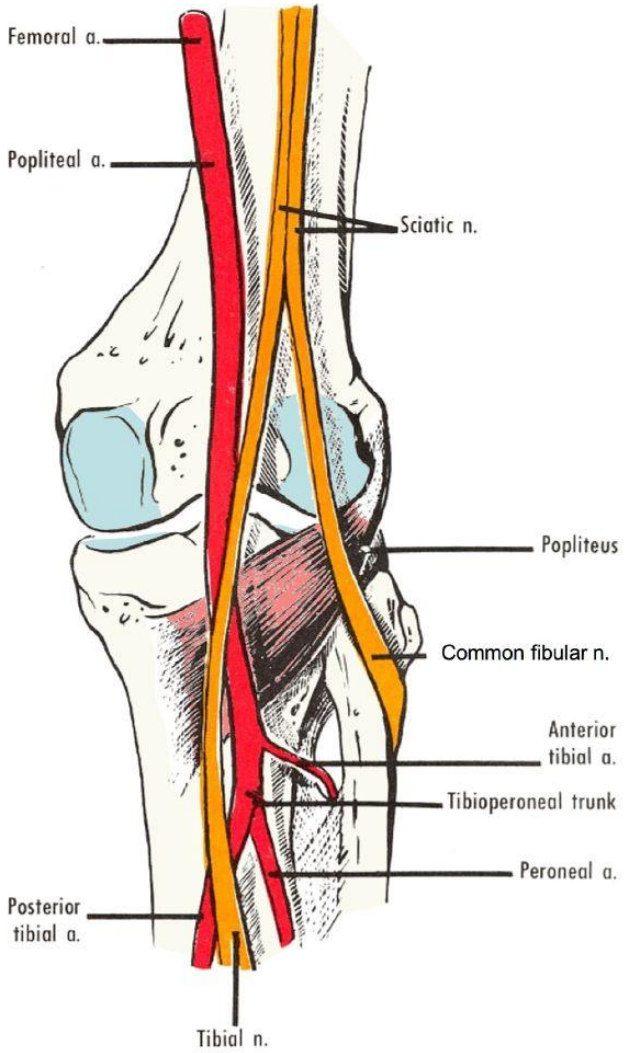
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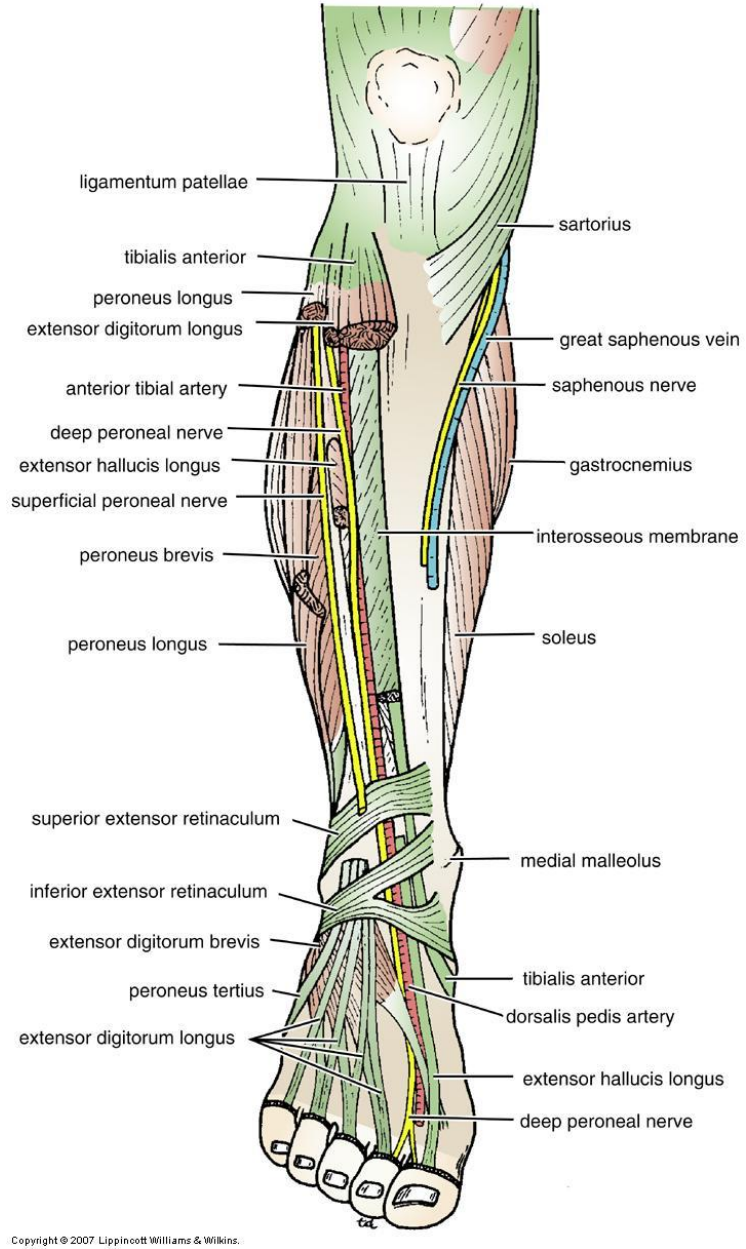


Anterior view

Anterior Tibial Artery

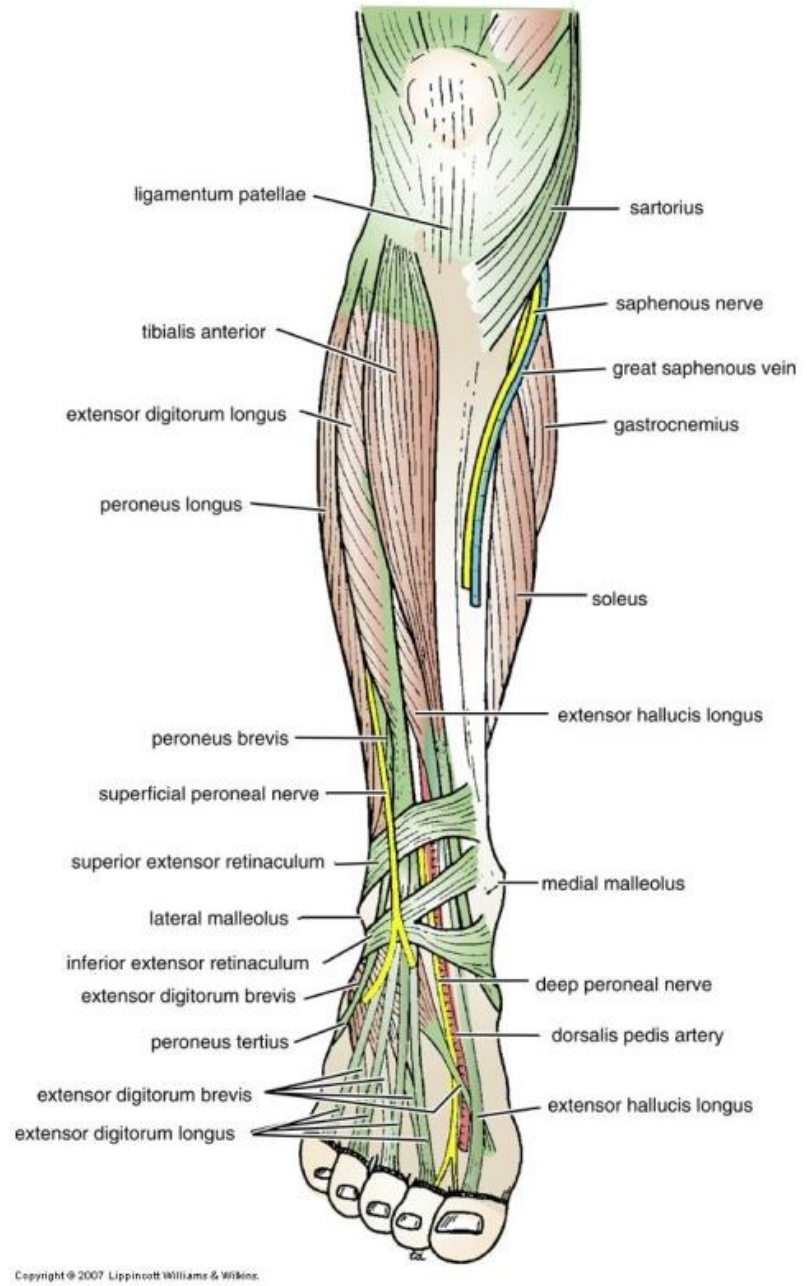
- Smaller than posterior tibial
- Begin at the lower border of the popliteus m.
- Traverse the interosseous membrane
- Companies the deep peroneal n.





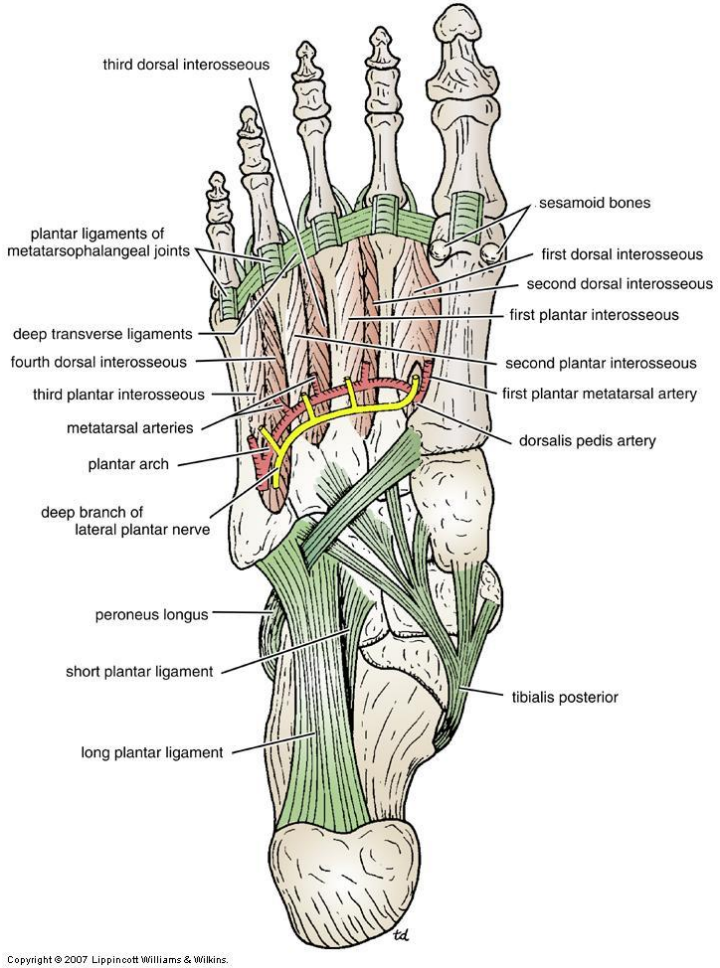
Anterior Tibial Artery

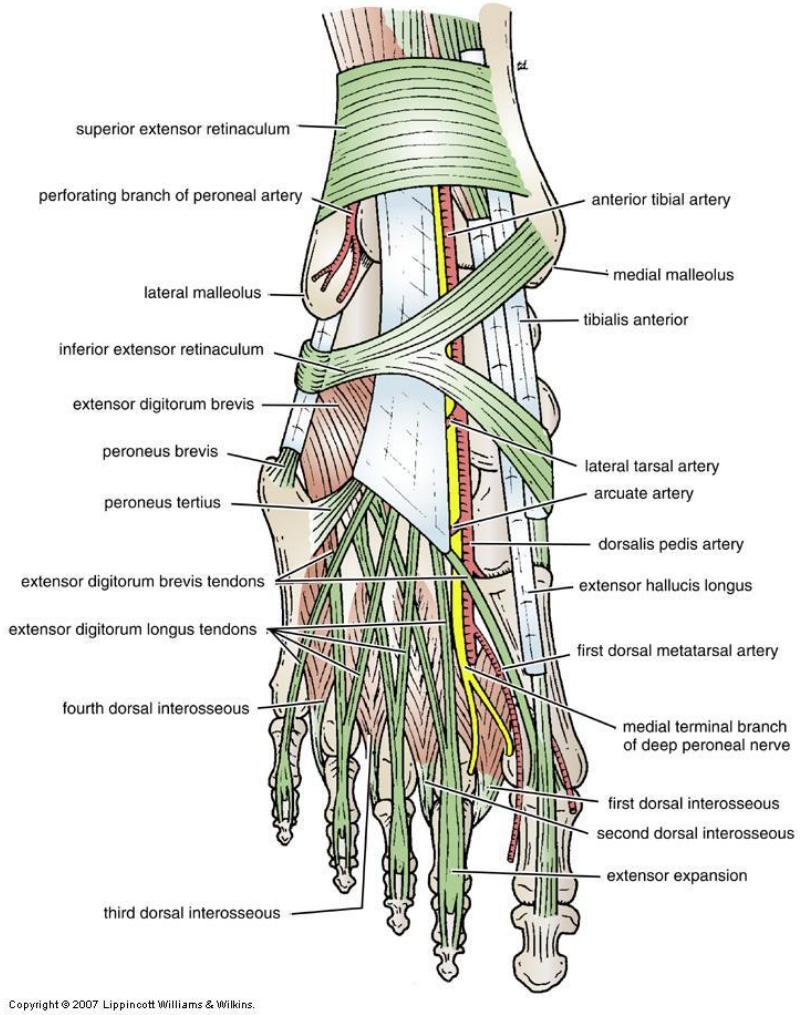
- Inferiorly
- Becomes superficial
- Bass deep to the extensor retinaculum
- Between tendons of extensor hallucis longus m. and extensor digitorum longus m.
- Branches
- Muscular branches
- Anastomotic branches (knee & ankle joints)
- At ankle becomes dorsalis pedis a.



Dorsalis Pedis Artery

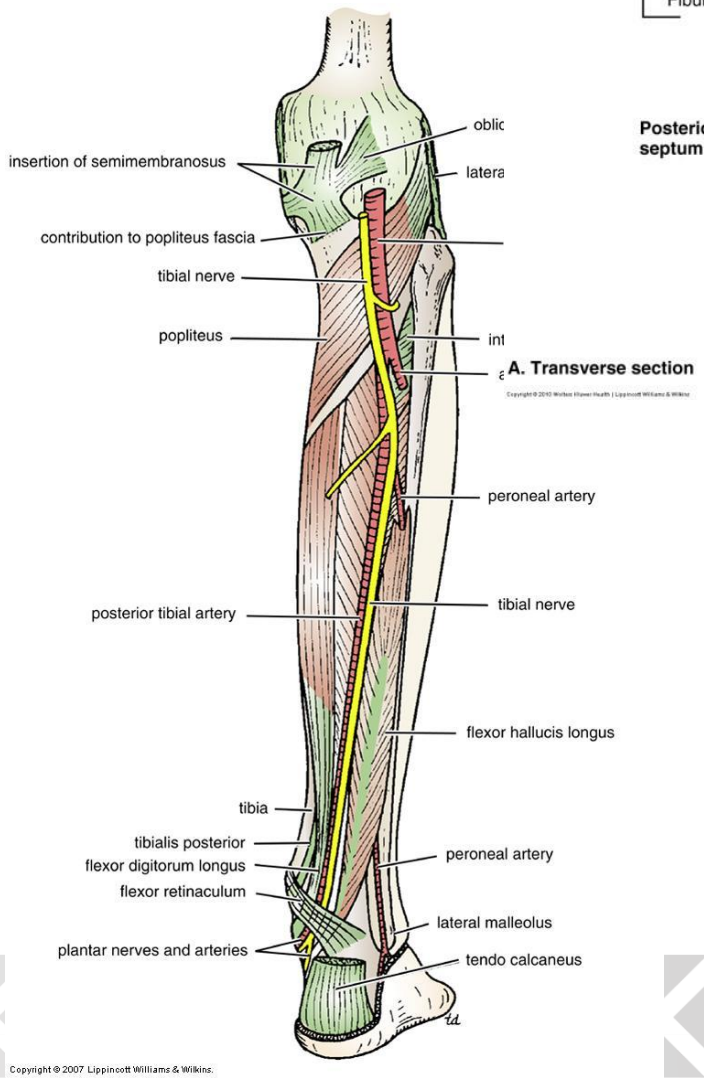
- Superficial
- Relations
  - Inferior extensor retinaculum
  - 1st tendon of extensor digitorum (hallucis) previs m.
  - 1st dorsal interosseous m.
- Branches
  - Lateral tarsal a.
  - Arcuate a.
    - Deep to the extensors
    - Metatarsal branches 2-5
  - 1st dorsal metatarsal a.
- At planter side join the planter arch

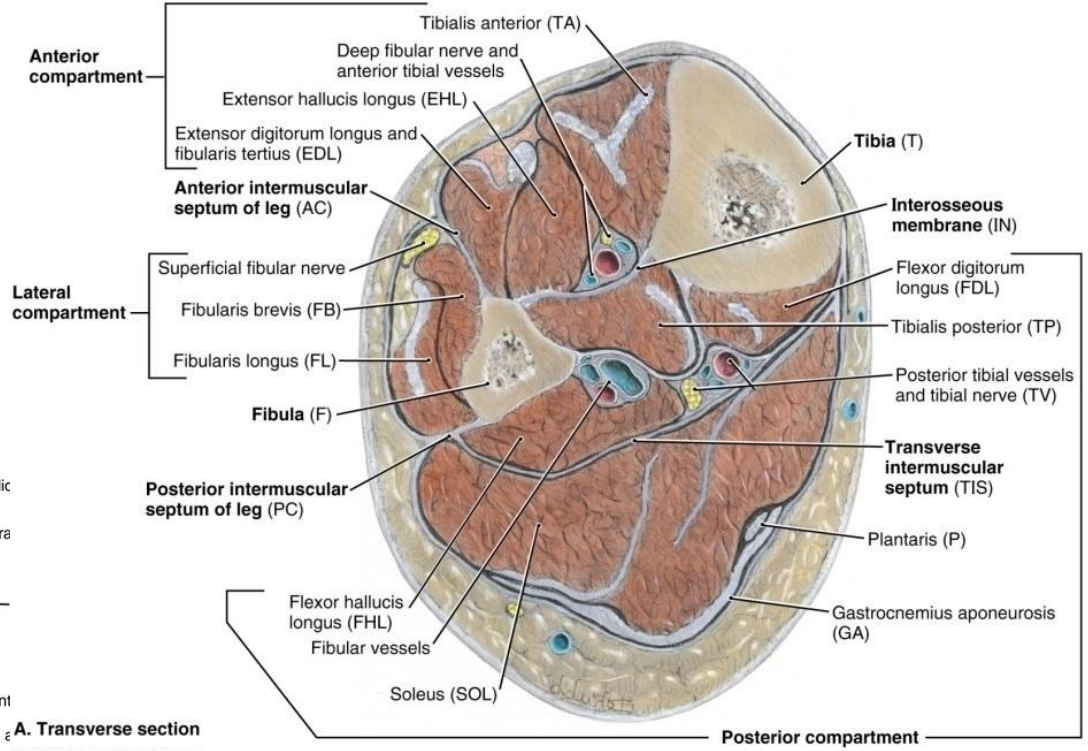




Posterior Tibial Artery

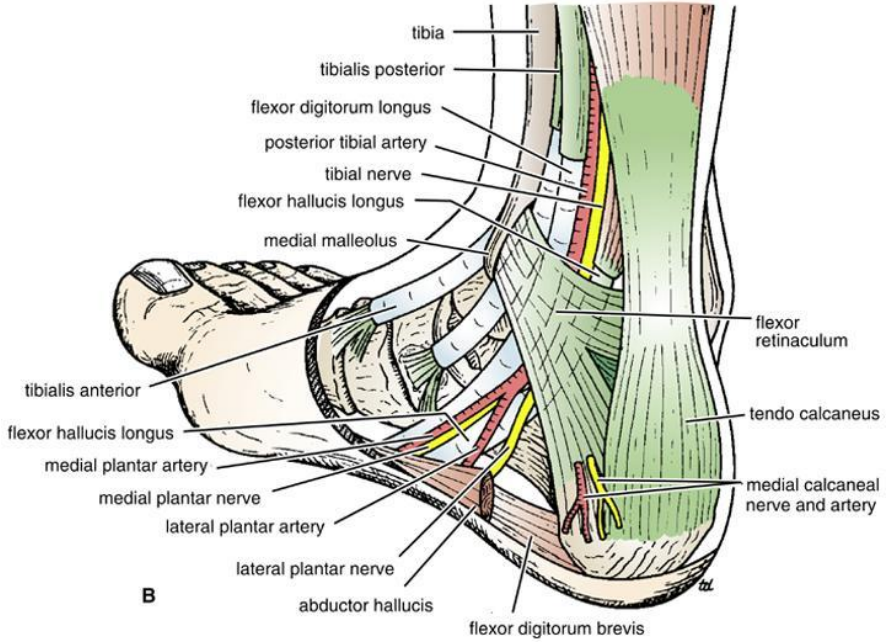
- Relations
  - Transverse intermuscular septum
  - Tibialis posterior
  - Inferiorly
  - Superficial
  - Deep to the flexor retinaculum
- Branches
  - Peroneal a.
    - Deep to the flexor hallucis longus m.
    - Branches
      - Muscular branches
      - Nutrient a. to fibula
      - Perforating branch
      - Traverse the interosseous membrane
  - Muscular branches
  - Nutrient a. to tibia
  - Anastomotic branches (ankle joint)
  - Medial and lateral planter aa.

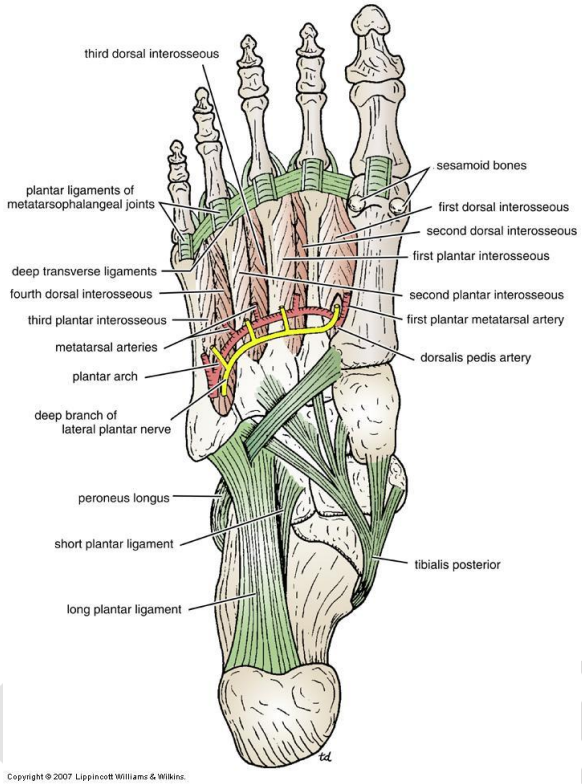


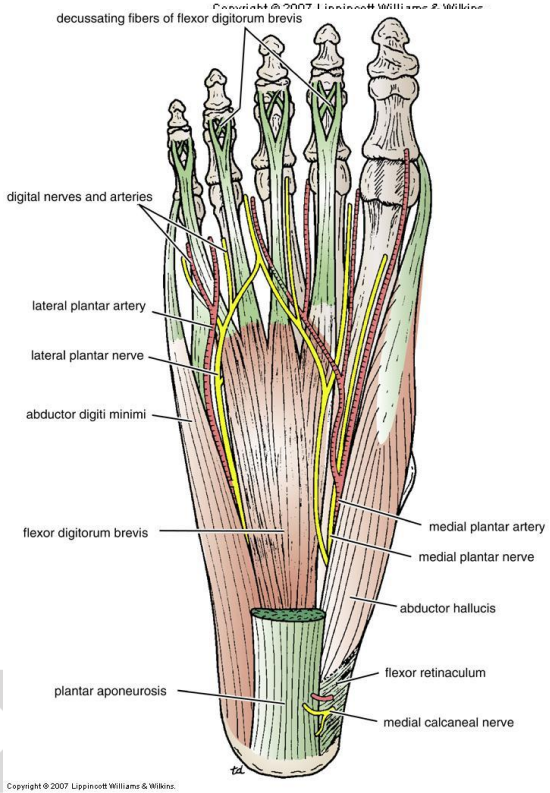


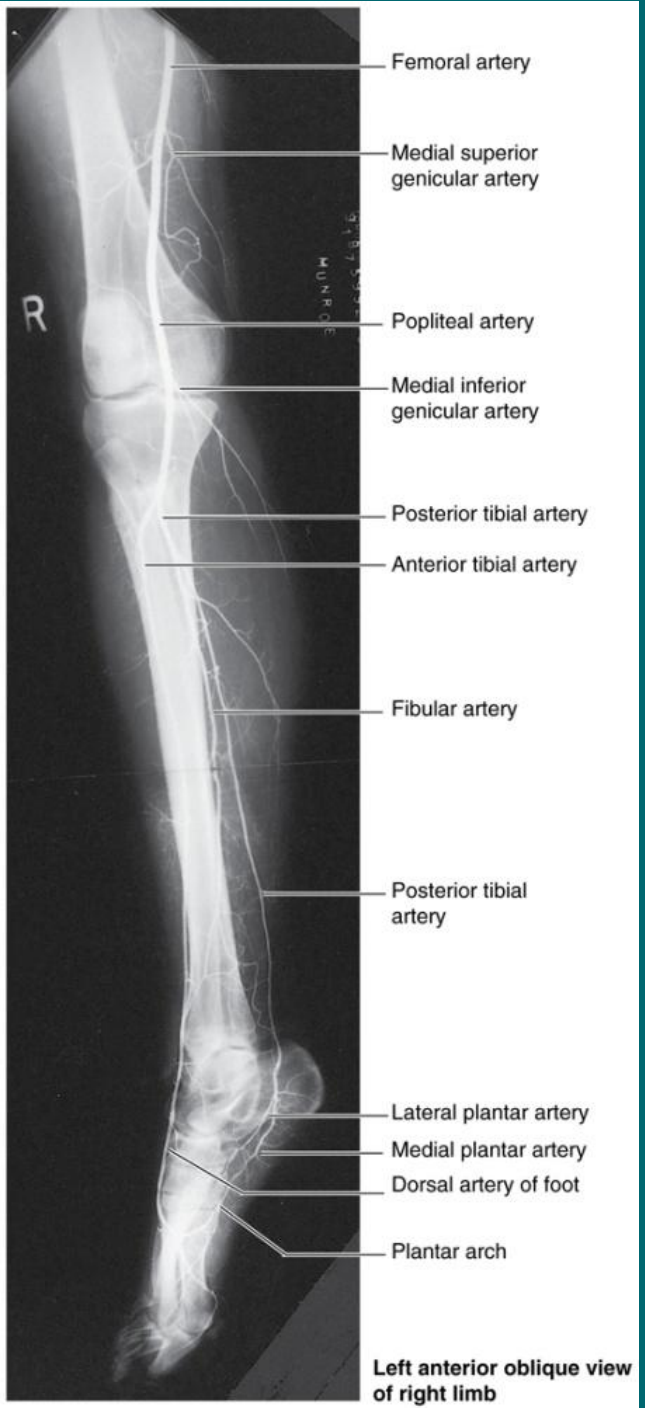
Posterior Tibial Artery: terminal branches

- Medial planter a.
  - Smaller
  - Deep to abductor hallucis m.
  - Branches
    - Muscular branches
    - Cutaneous branches
    - Articular branches
- Lateral planter a.
  - Deep to the abductor hallucis m. and flexor digitorum brevis m.
  - Continue as planter arch
    - Planter digital aa.
    - Branches
      - Muscular branches
      - Cutaneous branches
      - Articular branches





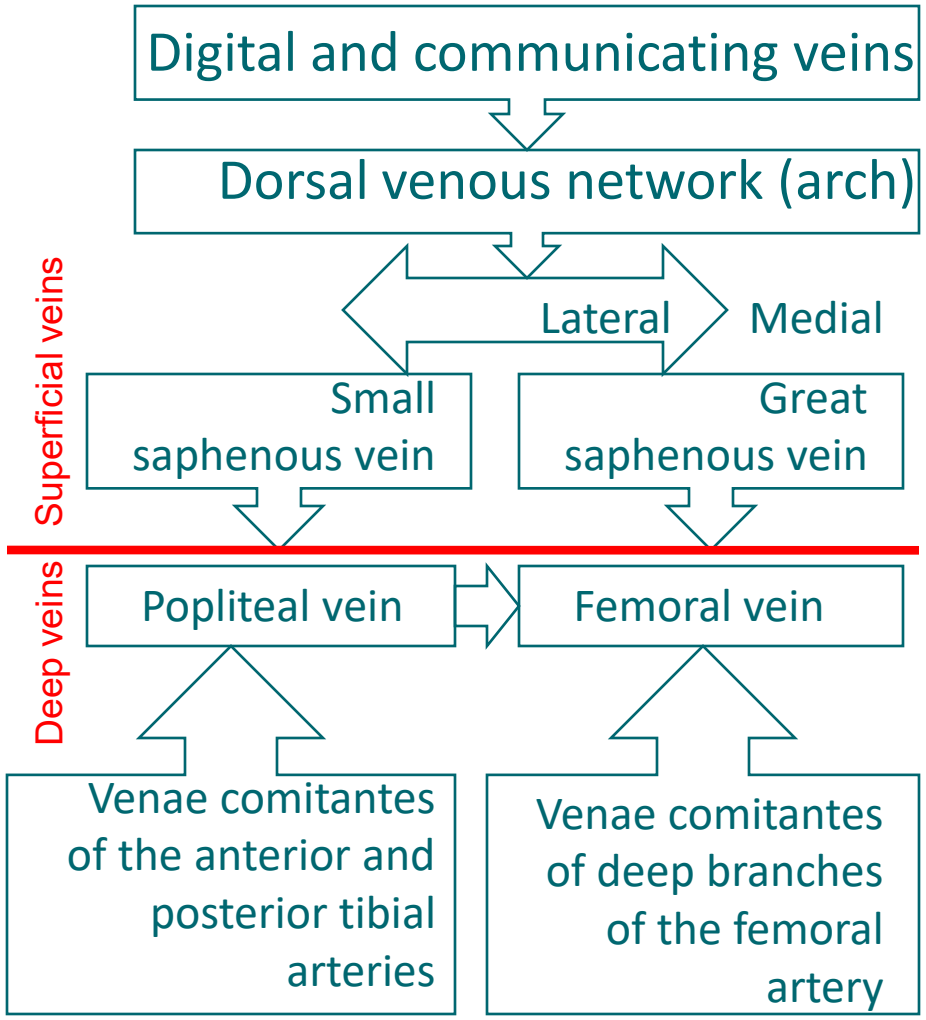


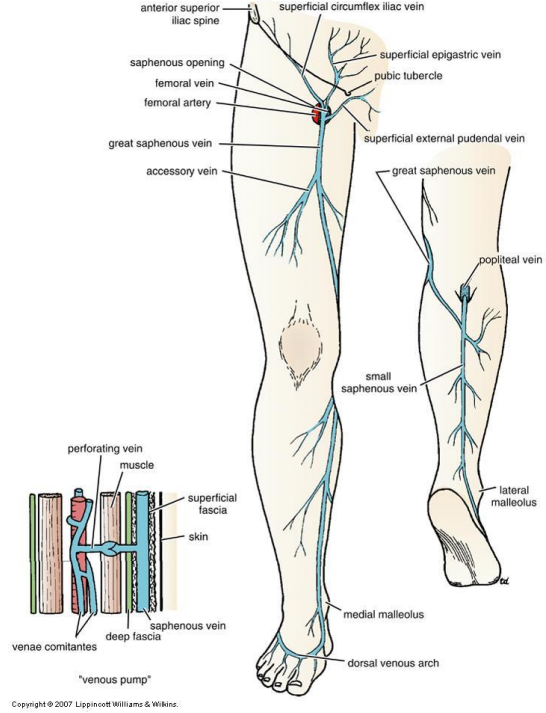


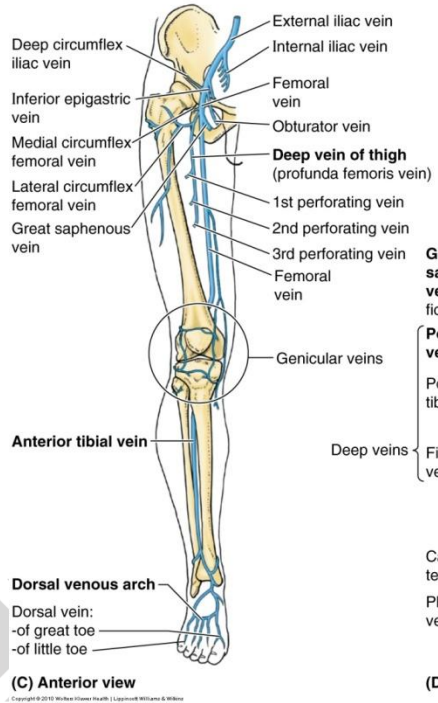
Left anterior oblique view of right limb

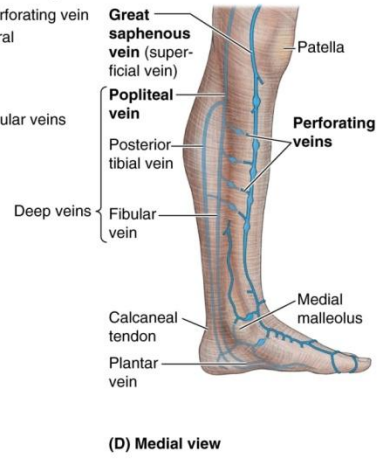
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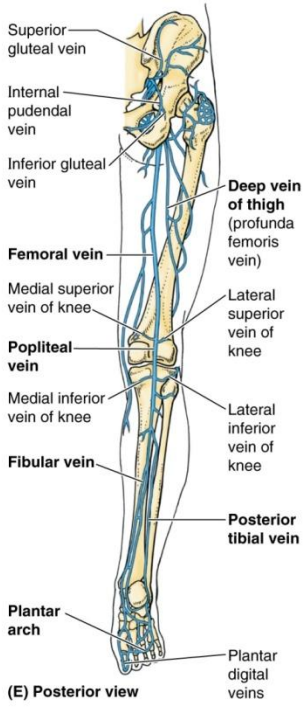
Veins of the Lower Limb





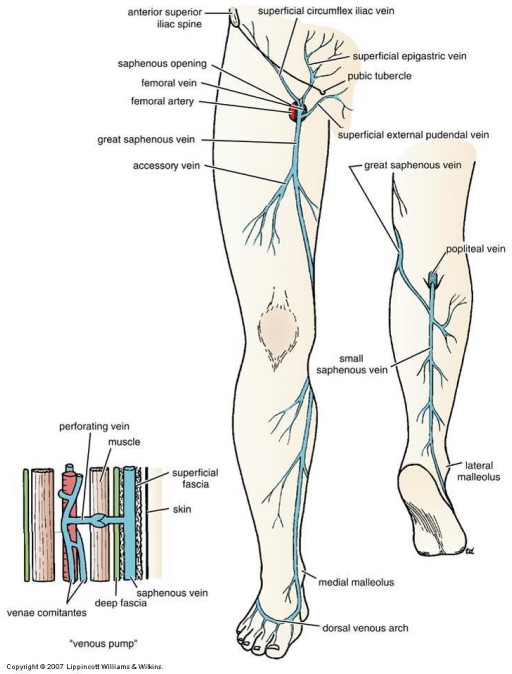


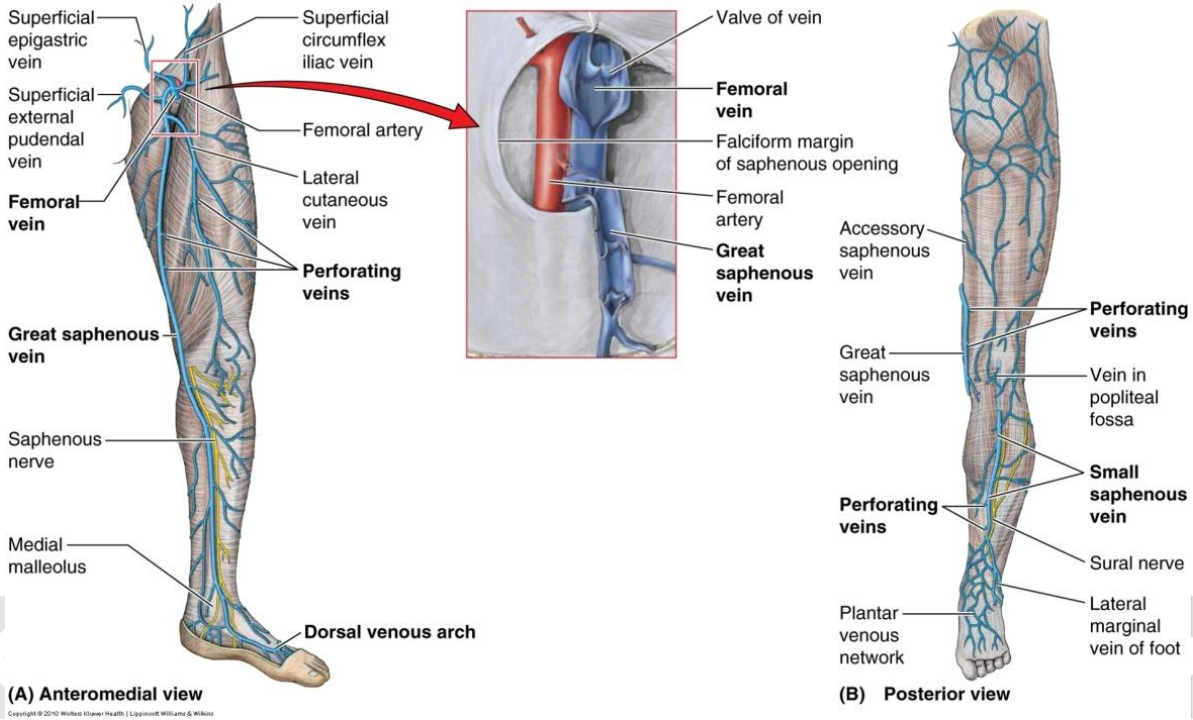




Veins of the Lower Limb

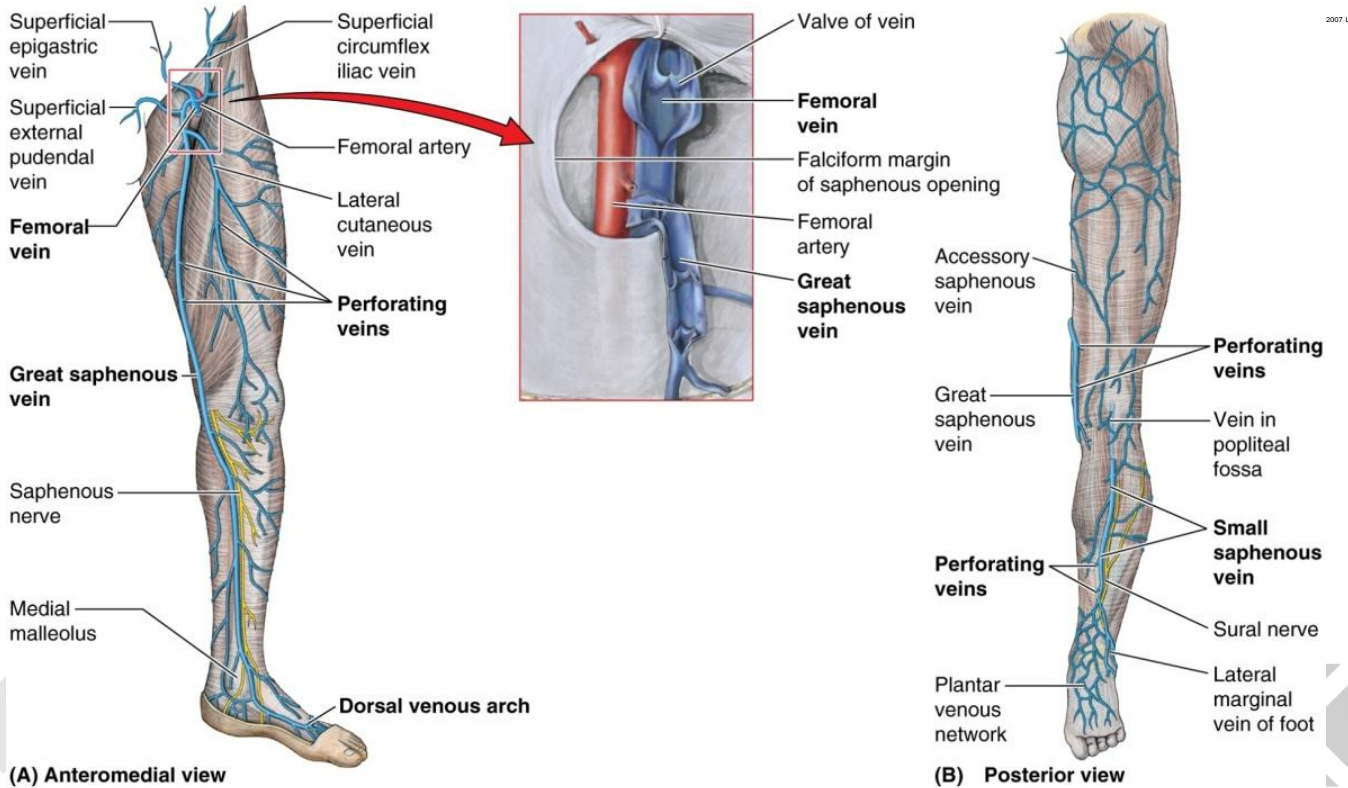
- Great saphenous vein
  - Medial side
  - Anterior to medial malleolus
  - Posterior to knee
  - Accompanies the saphenous n.
  - Connected to the small saphenous v. by anastomotic branches
  - Connected with deep vv. by perforating branches
  - Tributaries
    - Superficial circumflex iliac v.
    - Superficial epigastric v.
    - Superficial external pudendal v.
    - Accessory v.
    - Numerous other subcutaneous vv.

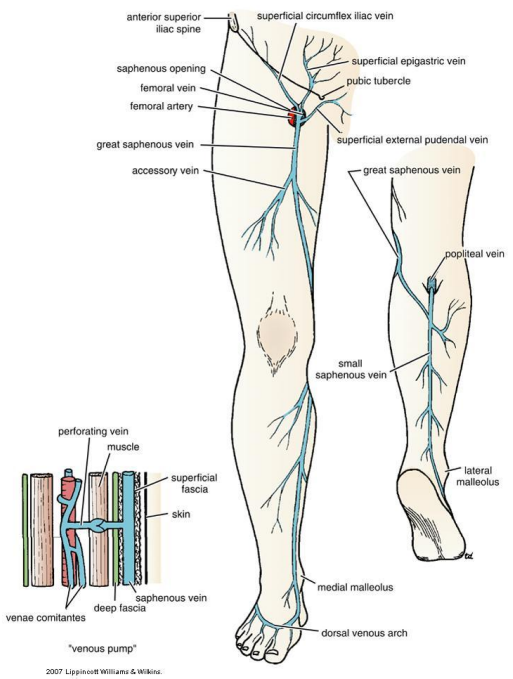




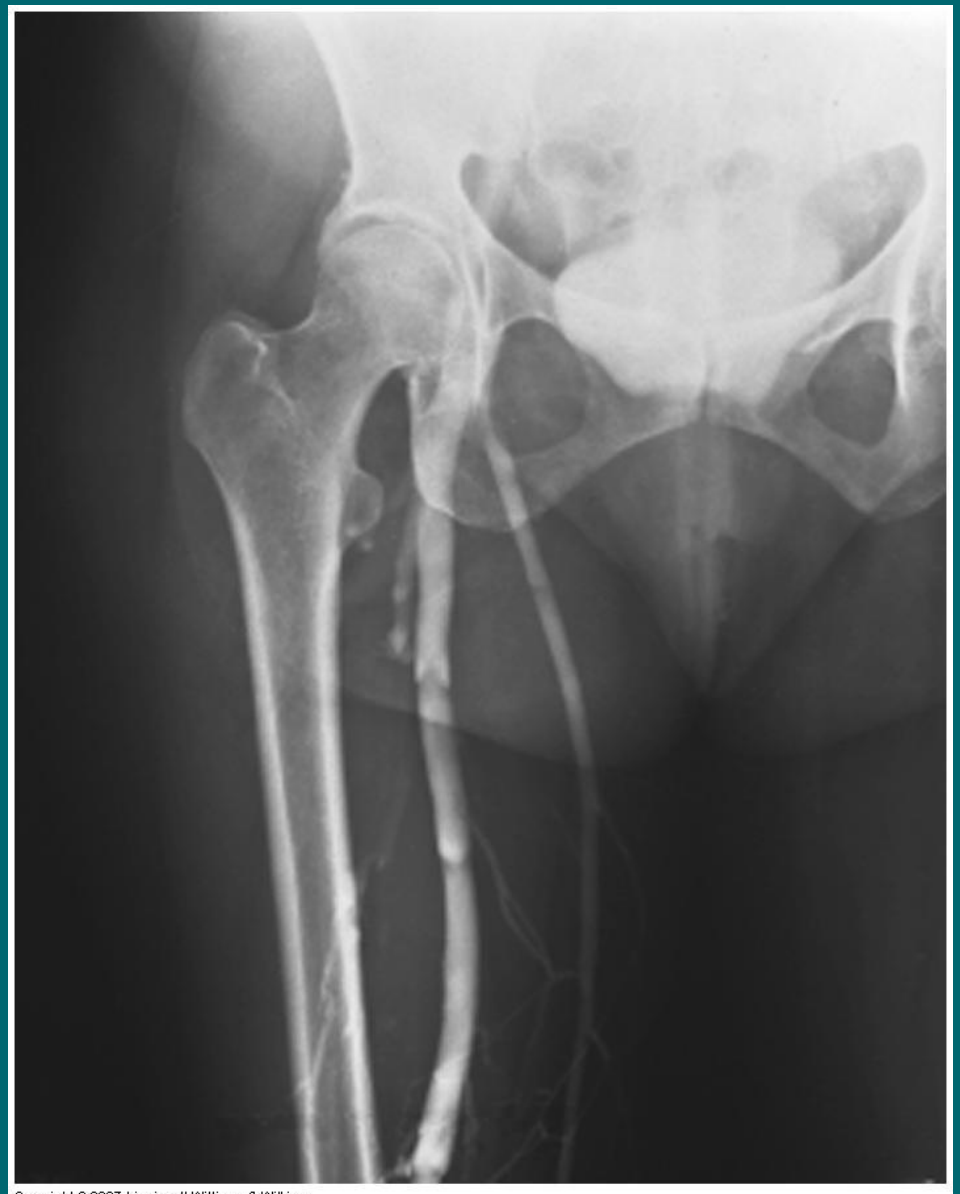
Veins of the Lower Limb

- Small saphenous vein
  - Lateral side
  - Posterior to lateral malleolus
  - Accompanies the sural n.
  - Lateral to the calcaneal tendon
  - Between the heads of the gastrocnemius m.
- Tributaries
  - Small subcutaneous vv.
  - Communicating veins with deep vv.
  - Anastomotic branches with the great saphenous v.





Venogram of Femoral Vein and its Tributaries



Sympathetic Innervation for Lower Limb Vessels

- Preganglionic sympathetic fibers rise from T10-L2 spinal segments
- Postganglionic fibers rise from the lumbar and sacral ganglia of sympathetic trunk
- Postganglionic fibers reach BV through the peripheral nerves of the lumbar and sacral plexuses. Examples:

Femoral a. ----- femoral & obturator nn.➢

Distal aa. (anterior & posterior tibial) ----- sciatic n. ➢ branches (common peroneal & tibial)

Great saphenous v. ----- saphenous n.➢

Small saphenous v. ----- sural n.➢

Lymphatic Drainage of Lower Limb

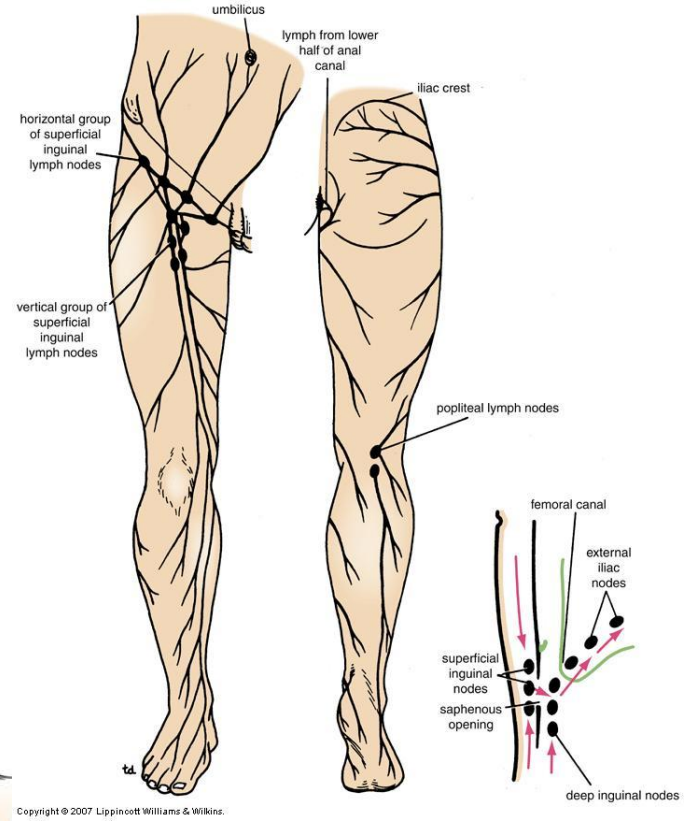
Superficial and deep lymphatic vessels follow their corresponding blood vessels

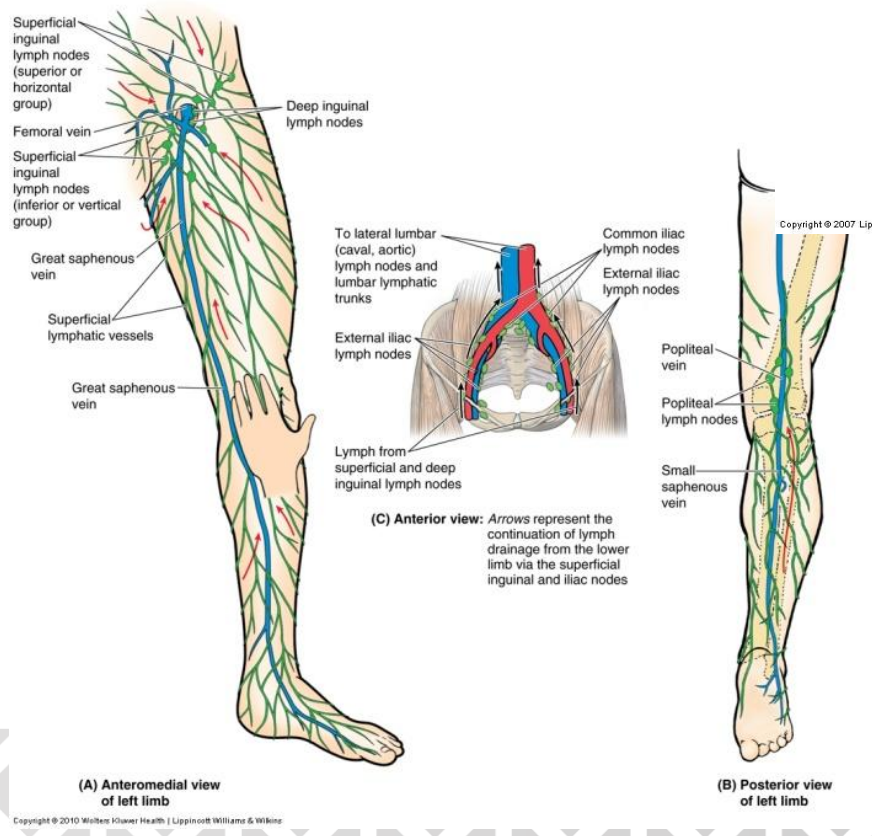
Popliteal lymph nodes •

Superficial vessels that accompany the small saphenous vein •

Lateral foot •

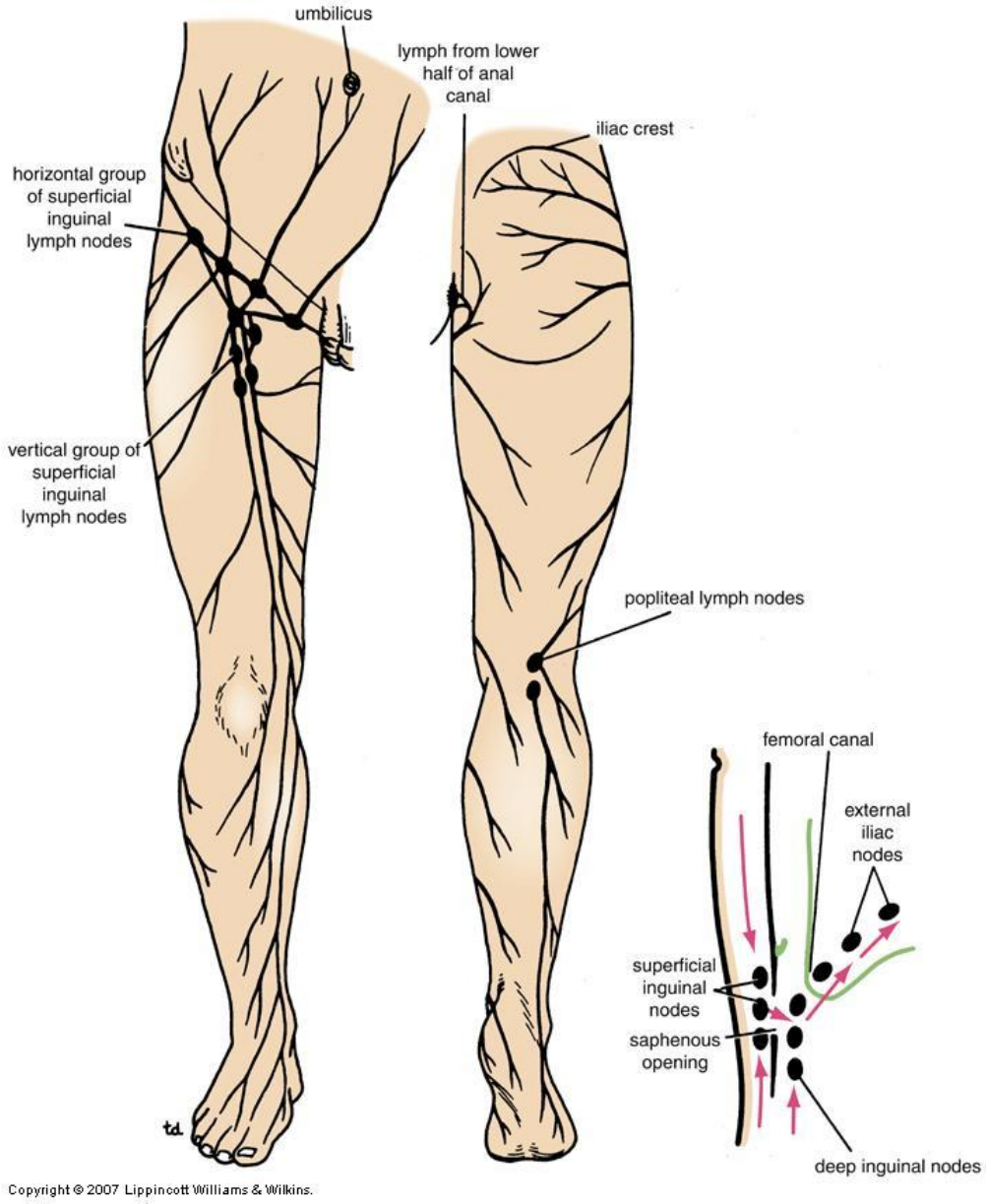
Lateral and posterior leg •





Lymphatic Drainage of Lower Limb

- Superficial inguinal lymph nodes
  - Horizontal group
    - Superficial lower anterior abdominal wall
    - Perineum
    - external genitalia
    - Buttock
  - Vertical group
    - Superficial vessels that accompany the great saphenous vein
- Deep inguinal lymph nodes
  - Superficial inguinal lymph nodes
  - Popliteal lymph nodes
  - Deep vessels
- External iliac nodes
  - Deep inguinal lymph nodes



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Thanks